

Ricardo Andrés Calvo Méndez

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SUMMARY

Ph.D. student in **software security** and **formal verification** (ACM CCS, FSE, ASE, MSR) with **3+ years of industry experience** in QA and full-stack development. Seeking **Summer 2027** opportunities.

EDUCATION

Ph.D. in Electrical and Computer Engineering, *Purdue University* 2025–2030 (expected)
B.Sc. in Systems and Computer Engineering, *Universidad Nacional de Colombia* 2018–2023

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Research Assistant Aug 2025–Present
Purdue University, West Lafayette, IN

- Co-develop AutoSOUP, an LLM-based system that automates memory-safety verification of C code (CCS 2026).
- Designed its **debugging agent**, which diagnoses failed proofs and proposes fixes; set up cluster evaluations.
- It exposed **66.7% of 60 known CVEs** (vs. 28.3% for a coding agent) and found **20 new vulnerabilities**.

Graduate Teaching Assistant, ECE 30861: Software Engineering Fall 2026
Purdue University, West Lafayette, IN

- Grade assignments and hold office hours for **~150 students**.

Test Engineer & QA Analyst Sep 2024–Jul 2025
Kravata SAS, Bogotá, Colombia

- As the **first and only QA engineer**, defined the company's testing strategy and standards from scratch.
- Automated unit, end-to-end and CI/CD tests for the **4 most critical services** with Cucumber and Selenium.

Full-Stack Software Developer Aug 2023–Aug 2024
Junior Software Developer Aug 2021–Jan 2023

Fit Ideas SAS, Bogotá, Colombia

- Led **4 developers** on a **white-label** fuel sales app: defined architecture, reviewed code, supported Scrum.
- Built web and mobile apps (Spring Boot, React, Django, Flutter) and a configurable public-data web scraper.

Undergraduate Visiting Scholar (UREP-C) Feb 2023–Jul 2023
Purdue University, West Lafayette, IN

- With mentorship, designed and implemented EmNetTest's fuzzing strategy and its evaluation (ASE 2023).
- Helped triage results: reproduced **12 known vulnerabilities** and found **7 new defects (6 CVEs)**.

SELECTED PUBLICATIONS

- P. Amusuo, **R. Calvo**, et al. *AutoSOUP: Safety-Oriented Unit Proof Generation for Component-level Memory-Safety Verification* [arXiv](#). ACM CCS 2026. Presenting author.
- P. Amusuo, D. Liu, **R. Calvo**, et al. *Lessons from Mitigating False Positives in Google's OSS-Fuzz-Gen* [DOI](#). FSE Industry 2026. Presenting author.
- H. Peng, A.Z. Qiu, **R. Calvo**, et al. *How Do Agents Perform Code Optimization? An Empirical Study* [DOI](#). MSR Mining Challenge 2026. Presenting author.
- P. Amusuo, **R. Calvo**, Z. Xu, A. Machiry, and J.C. Davis. *Systematically Detecting Packet Validation Vulnerabilities in Embedded Network Stacks* [DOI](#). ASE 2023.

SKILLS

Research & Security: Formal verification, model checking (CBMC), program analysis, fuzzing, performance optimization

Programming: C, C++, Python, Java, JavaScript, TypeScript, Dart, SQL

AI for SE: LLM agents, agentic pipelines, empirical evaluation

Engineering: Spring Boot, React, Django, Flask, Flutter, PostgreSQL, Selenium, Cucumber, CI/CD, Docker, Linux, Git

GRANTS

ACM CCS 2026 Student Conference Grant | ICSE 2026 Student Mentoring Workshop Grant (NSF, ACM SIGSOFT)